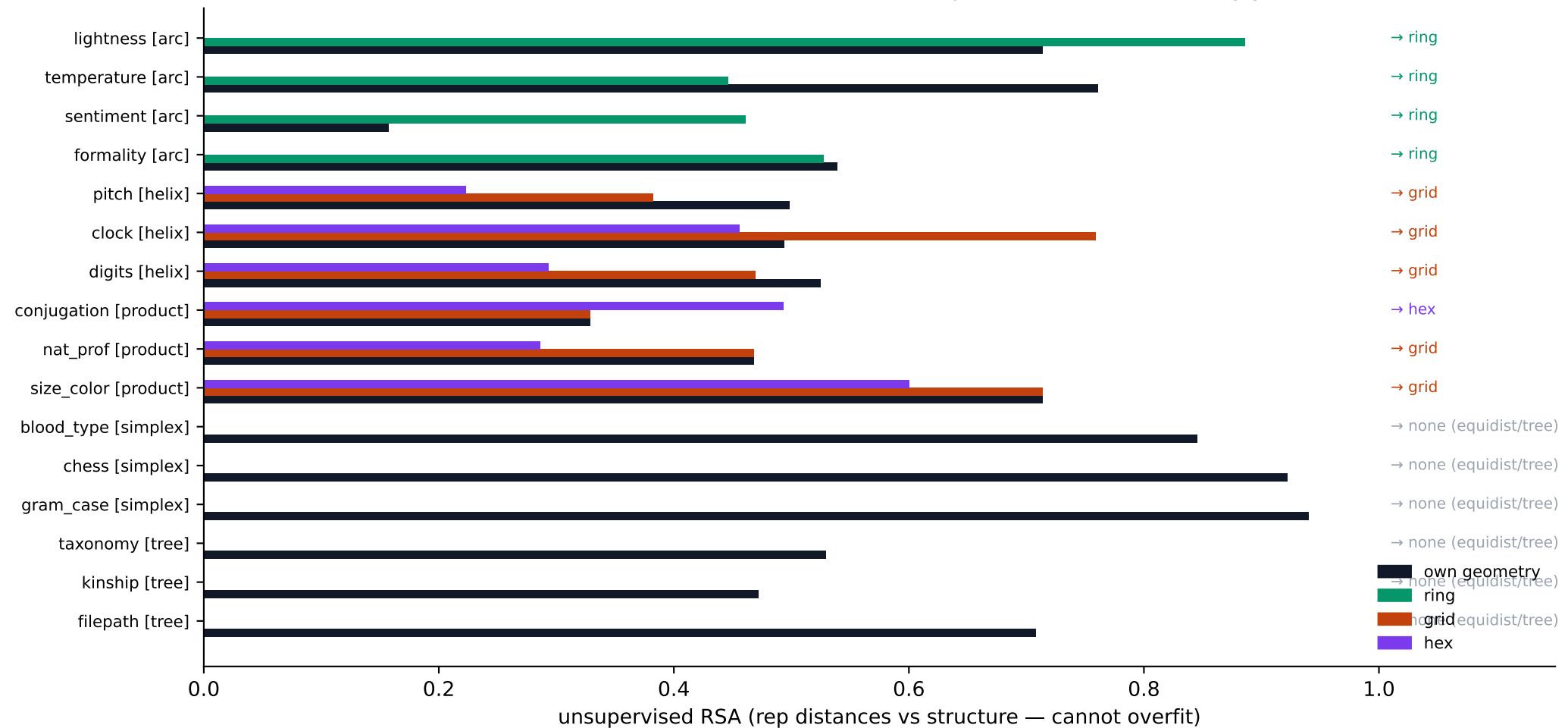


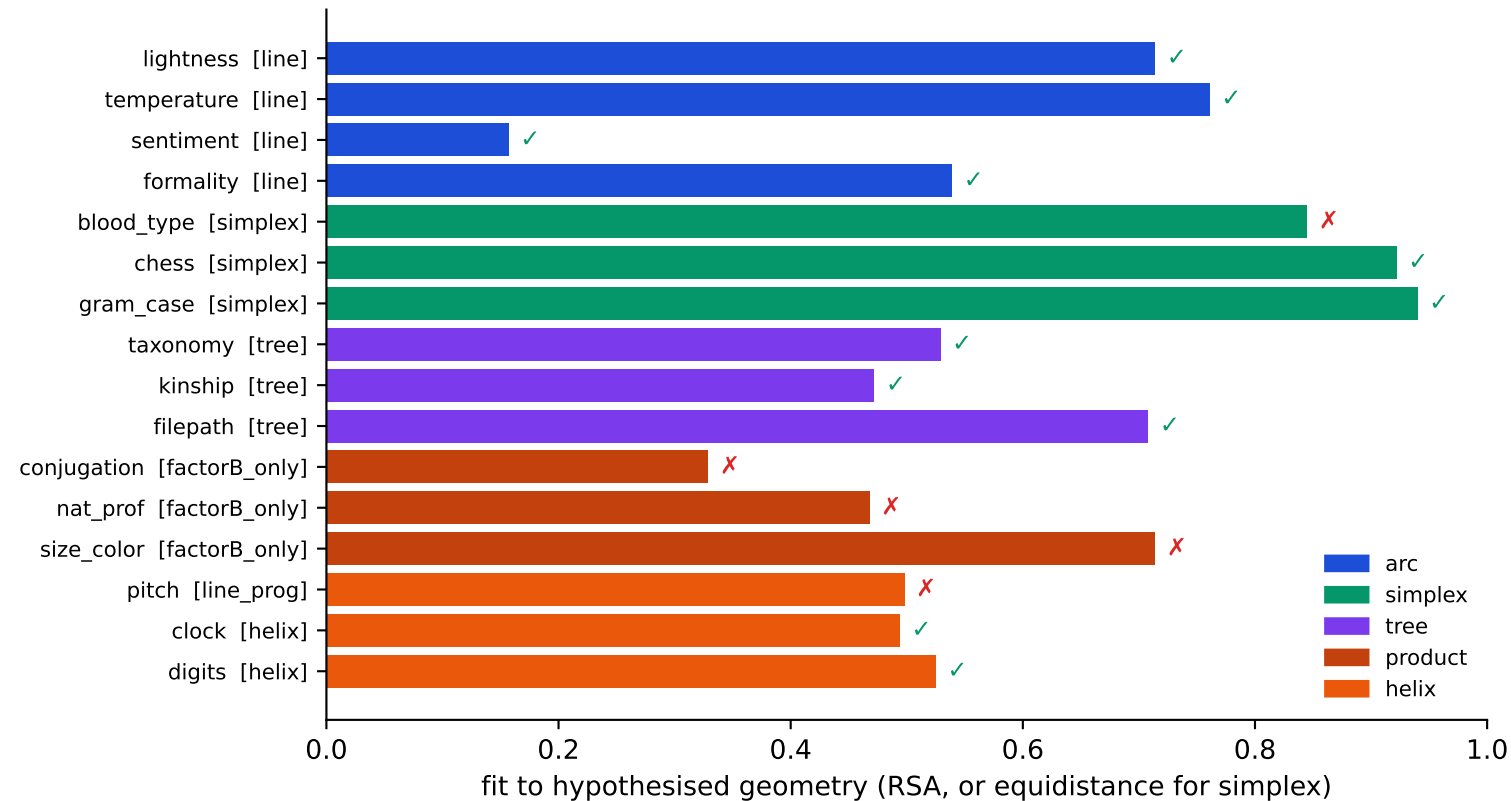
Does each concept family resemble the ring / grid / hex graph structures? (Llama)

Products & helices → GRID; some arcs → RING; simplices & trees → none of ring/grid/hex

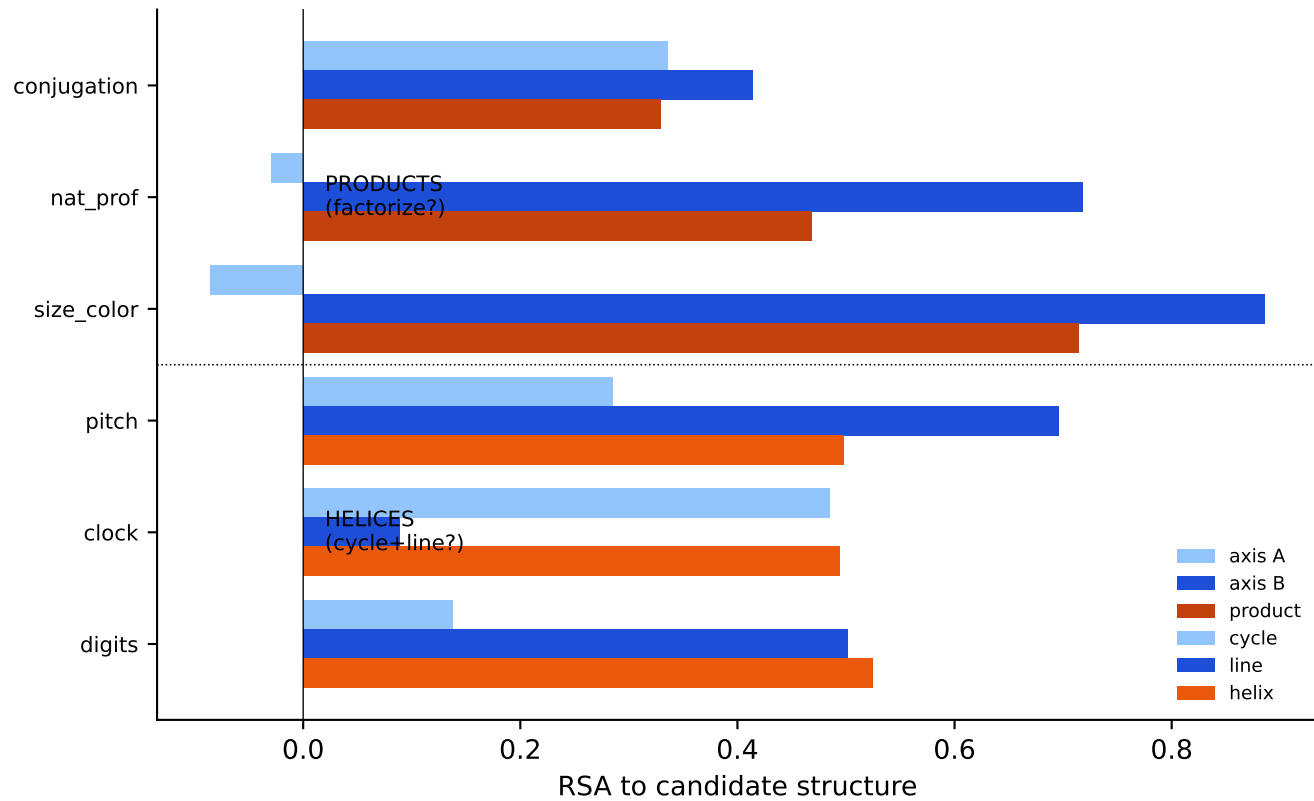


Geometry battery — FIT stage (Llama): which shape does the model use for each concept space?

does each concept space match its hypothesised shape?

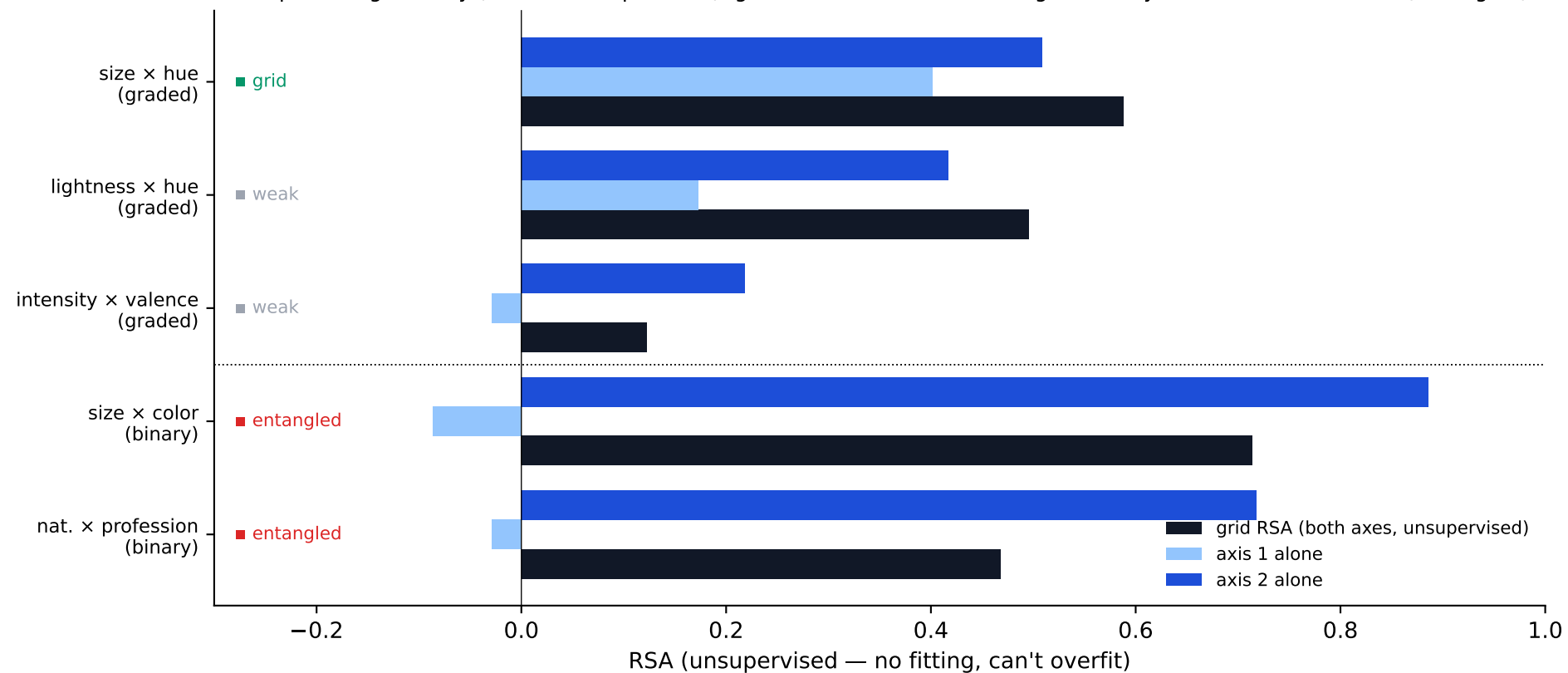


products: one axis dominates (entangled); helices: cycle vs line balance

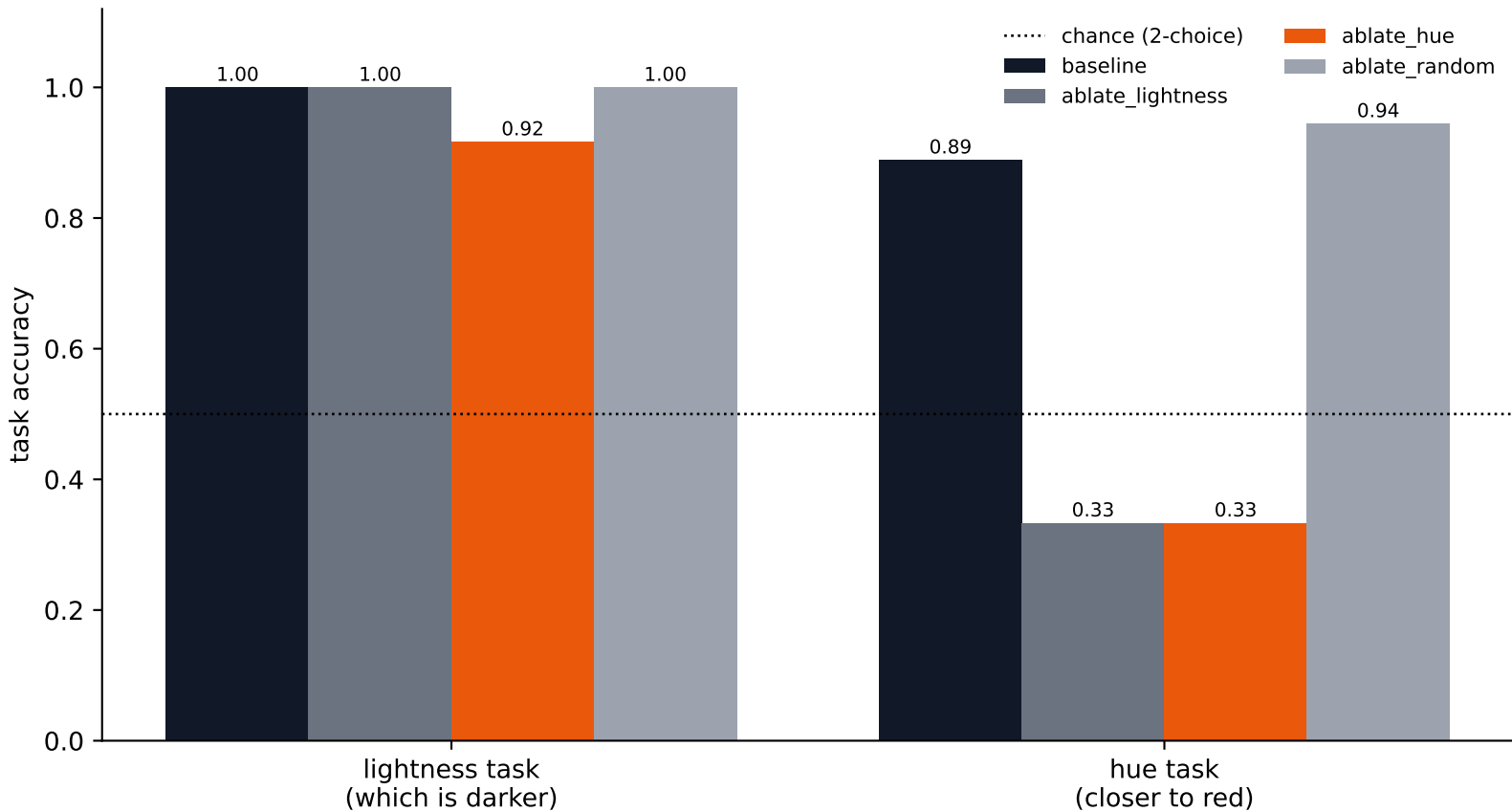


When does composition invoke a grid? (Llama) — balance of the two axes is the signature

Composition geometry (honest, unsupervised): graded axes → balanced 2D grid; binary axes → one dominates (entangled)

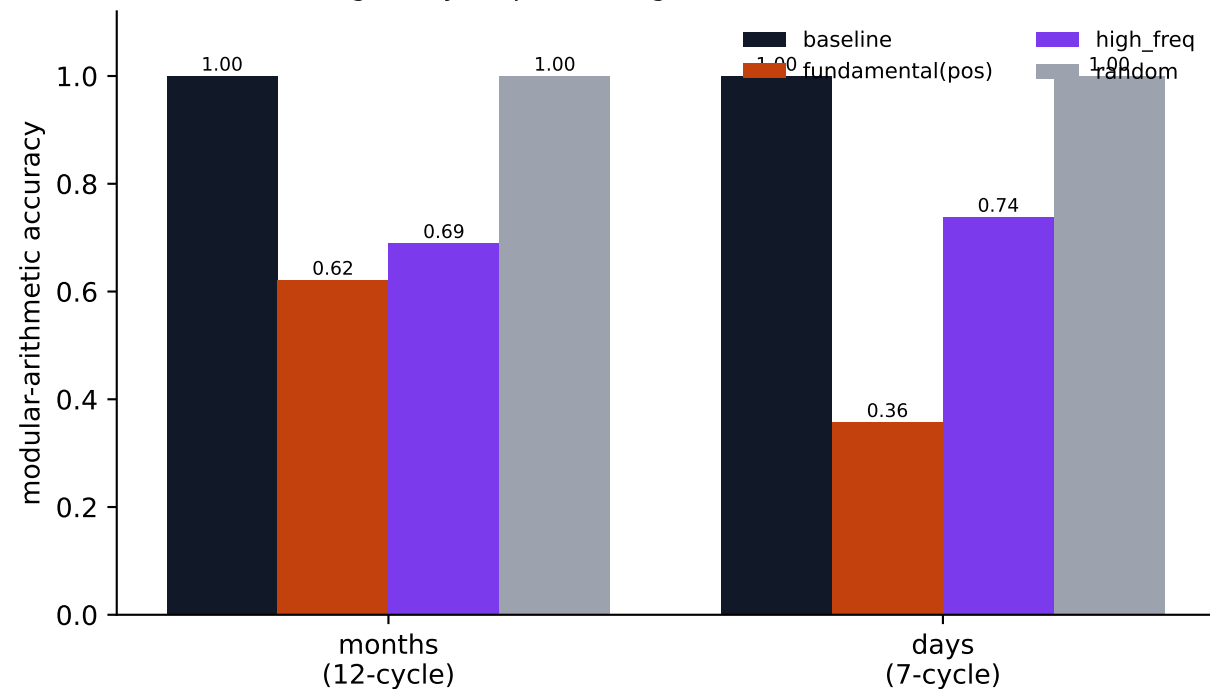


Colour-grid causal ablation (Llama) — LOO decode R^2 : lightness=0.97, hue=1.00
hue breaks under BOTH axis ablations (not random) $\Rightarrow$ axes non-orthogonal; lightness task is lexically trivial



Cyclic-arithmetic ablation (Llama): the model computes 'N months/days after X' through the circular-position eigenmode

ablating the cyclic-position eigenmode breaks the arithmetic



fundamental (m1/m2) dominates the representation

